



DEADLOCKS

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UNIT-II

DEADLOCKS: System model; Deadlock characterization, Methods for handling deadlocks, Deadlock prevention, avoidance, detection, and recovery from deadlock.

(Text 2: Chapter 8)

MAIN MEMORY: Paging, Structure of page table, Swapping. (Text 2: Chapter 9)

VIRTUAL MEMORY: Demand paging, Copy-on-write, Page replacement, Allocation of frames, Thrashing, Allocating Kernel Memory. (Text 2: Chapter 10)

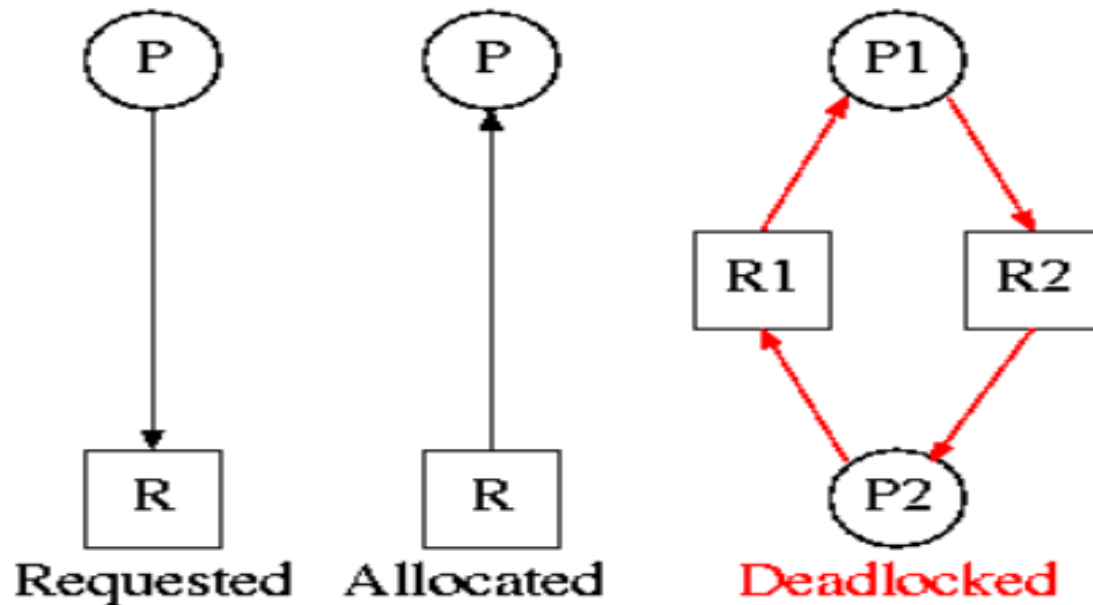
15 Hours

The Deadlock Problem

- If two or more processes are waiting on happening of some event, which never happens -> **Deadlock**
- A set of blocked processes each holding a resource and waiting to acquire a resource held by another process in the set.
- Example
 - System has 2 disk drives
 - P_1 and P_2 each hold one disk drive and each needs another one

The Deadlock Problem

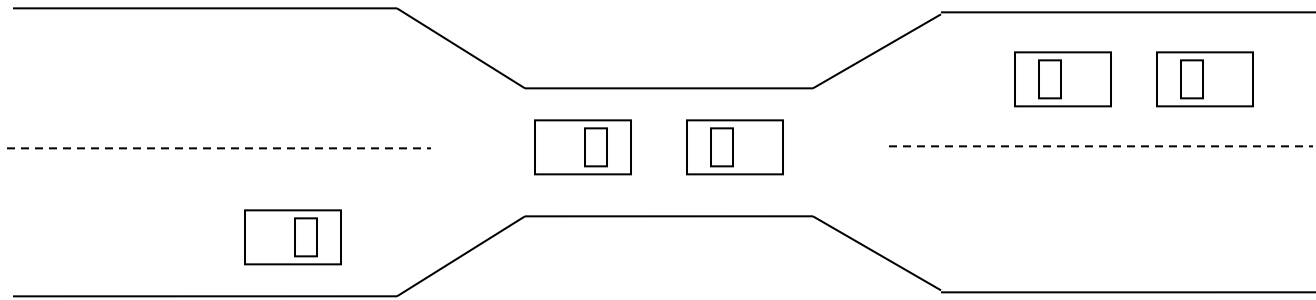
- A **deadlock** occurs when a every member of a set of processes is waiting for an event that can only be caused by a member of the set.
- Often the event waited for is the release of a resource



Deadlock



Bridge Crossing Example



- Traffic only in one direction
- Each section of a bridge can be viewed as a resource
- If a deadlock occurs, it can be resolved if one car backs up (preempt resources and rollback)
- Several cars may have to be backed up if a deadlock occurs
- Starvation is possible
- Note – Most OSes do not prevent or deal with deadlocks

System Model

- Resource types $R_1, R_2, \dots, R_m$
CPU cycles, memory space, I/O devices
- Each resource type R_i has W_i instances
- Each process utilizes a resource as follows:
 - **request**
 - **use**
 - **release**

Deadlock Characterization

Deadlock can arise if four conditions hold simultaneously.

- **Mutual exclusion:** only one process at a time can use a resource
- **Hold and wait:** a process holding at least one resource is waiting to acquire additional resources held by other processes
- **No preemption:** a resource can be released only voluntarily by the process holding it, after that process has completed its task
- **Circular wait:** there exists a set $\{P_0, P_1, \dots, P_n\}$ of waiting processes such that P_0 is waiting for a resource that is held by P_1 , P_1 is waiting for a resource that is held by P_2 , ..., P_{n-1} is waiting for a resource that is held by P_n , and P_n is waiting for a resource that is held by P_0

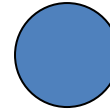
Resource-Allocation Graph

A set of vertices V and a set of edges E

- V is partitioned into two types:
 - $P = \{P_1, P_2, \dots, P_n\}$, the set consisting of all the processes in the system
 - $R = \{R_1, R_2, \dots, R_m\}$, the set consisting of all resource types in the system
- **request edge** – directed edge $P_i \rightarrow R_j$
- **assignment edge** – directed edge $R_j \rightarrow P_i$

Resource-Allocation Graph (Cont.)

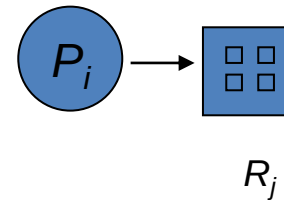
- Process



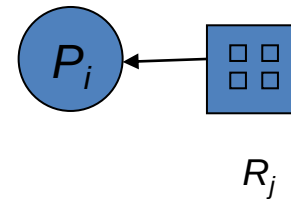
- Resource Type with 4 instances



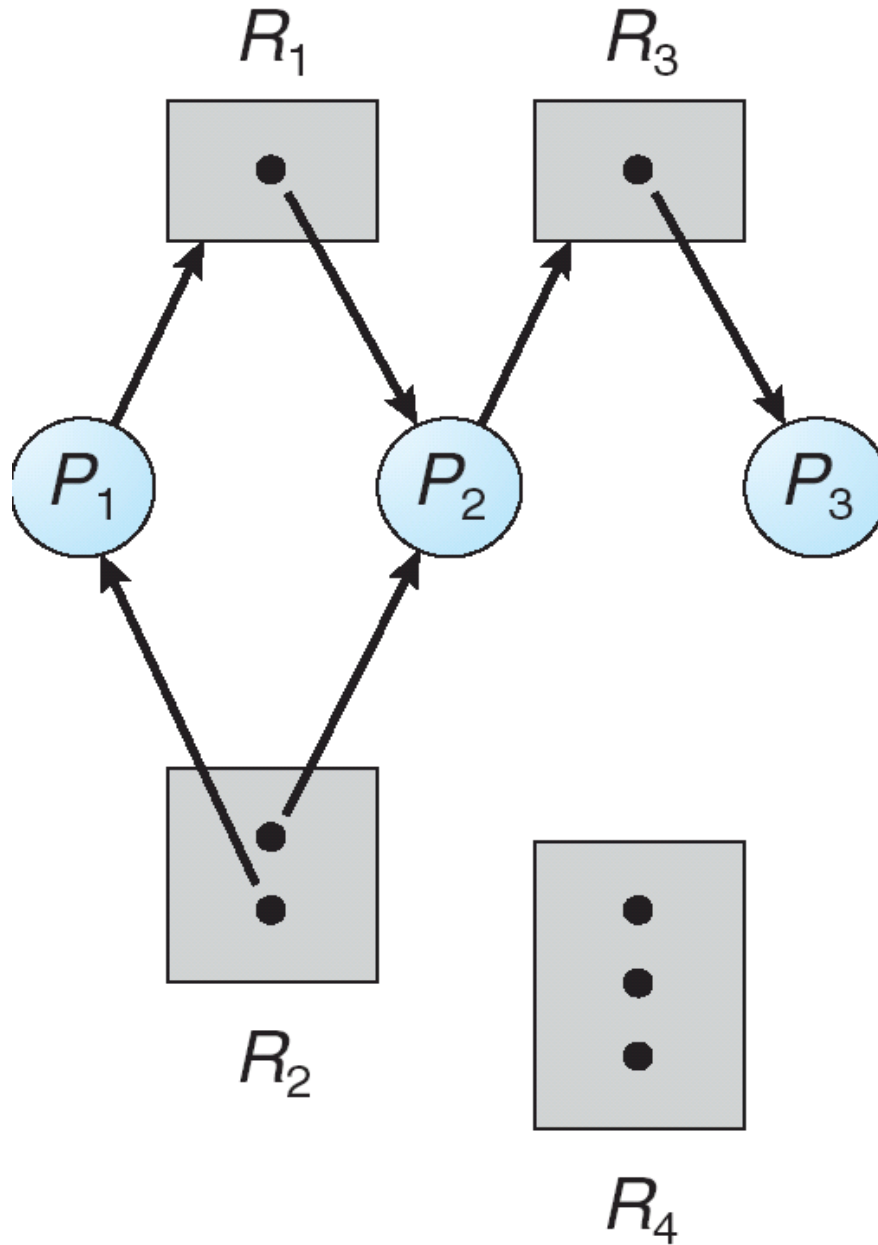
- P_i requests instance of R_j



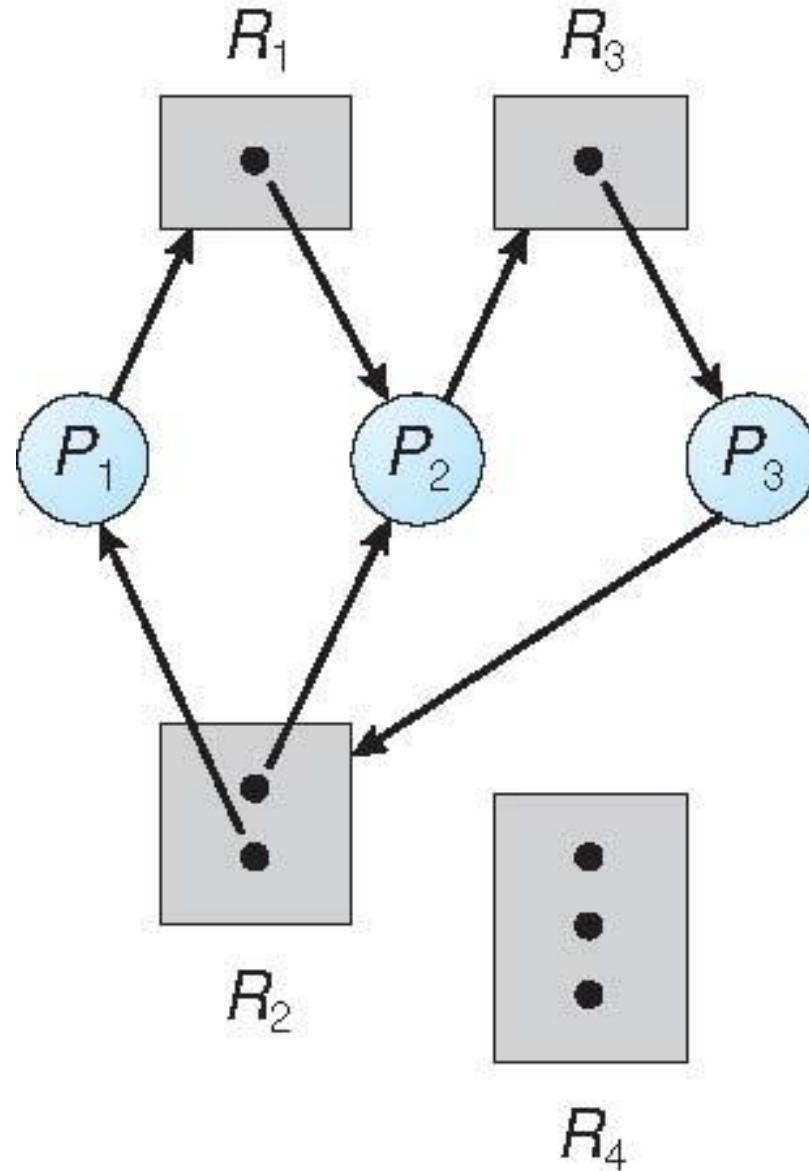
- P_i is holding an instance of R_j



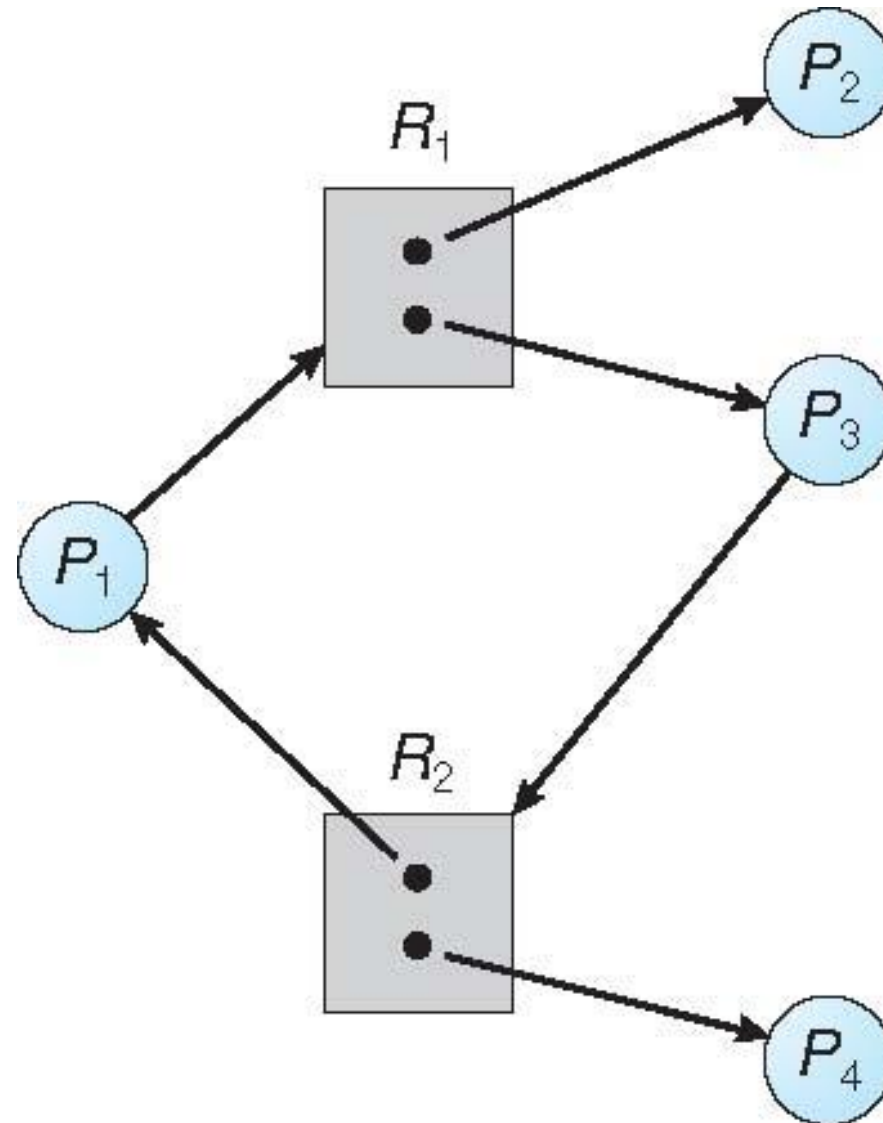
Example of a Resource Allocation Graph



Resource Allocation Graph With A Deadlock



Graph With A Cycle But No Deadlock



Basic Facts

- If graph contains no cycles $\Rightarrow$ no deadlock
- If graph contains a cycle $\Rightarrow$
 - if only one instance per resource type, then deadlock
 - if several instances per resource type, possibility of deadlock

Methods for Handling Deadlocks

- Ensure that the system will *never* enter a deadlock state
- Allow the system to enter a deadlock state and then recover
- Ignore the problem and pretend that deadlocks never occur in the system; used by most operating systems, including UNIX

Deadlock Prevention

Restrain the ways request can be made

- **Mutual Exclusion** – not required for sharable resources; must hold for nonsharable resources
- **Hold and Wait** – must guarantee that whenever a process requests a resource, it does not hold any other resources
 - Require process to request and be allocated all its resources before it begins execution, or allow process to request resources only when the process has none
 - Low resource utilization; starvation possible

Deadlock Prevention (Cont.)

- **No Preemption** –
 - If a process that is holding some resources requests another resource that cannot be immediately allocated to it, then all resources currently being held are released
 - Preempted resources are added to the list of resources for which the process is waiting
 - Process will be restarted only when it can regain its old resources, as well as the new ones that it is requesting
- **Circular Wait** – impose a total ordering of all resource types, and require that each process requests resources in an increasing order of enumeration

Deadlock Avoidance

Requires that the system has some additional *a priori* information available.

- Simplest and most useful model requires that each process declare the *maximum number* of resources of each type that it may need.
- The deadlock-avoidance algorithm dynamically examines the resource-allocation state to ensure that there can never be a circular-wait condition.
- Resource-allocation *state* is defined by the number of available and allocated resources, and the maximum demands of the processes.

Safe State

- When a process requests an available resource, system must decide if immediate allocation leaves the system in a safe state. System is in **safe state** if there exists a sequence $\langle P_1, P_2, \dots, P_n \rangle$ of ALL the processes is the systems such that for each P_i , the resources that P_i can still request can be satisfied by currently available resources + resources held by all the P_j , with $j < i$

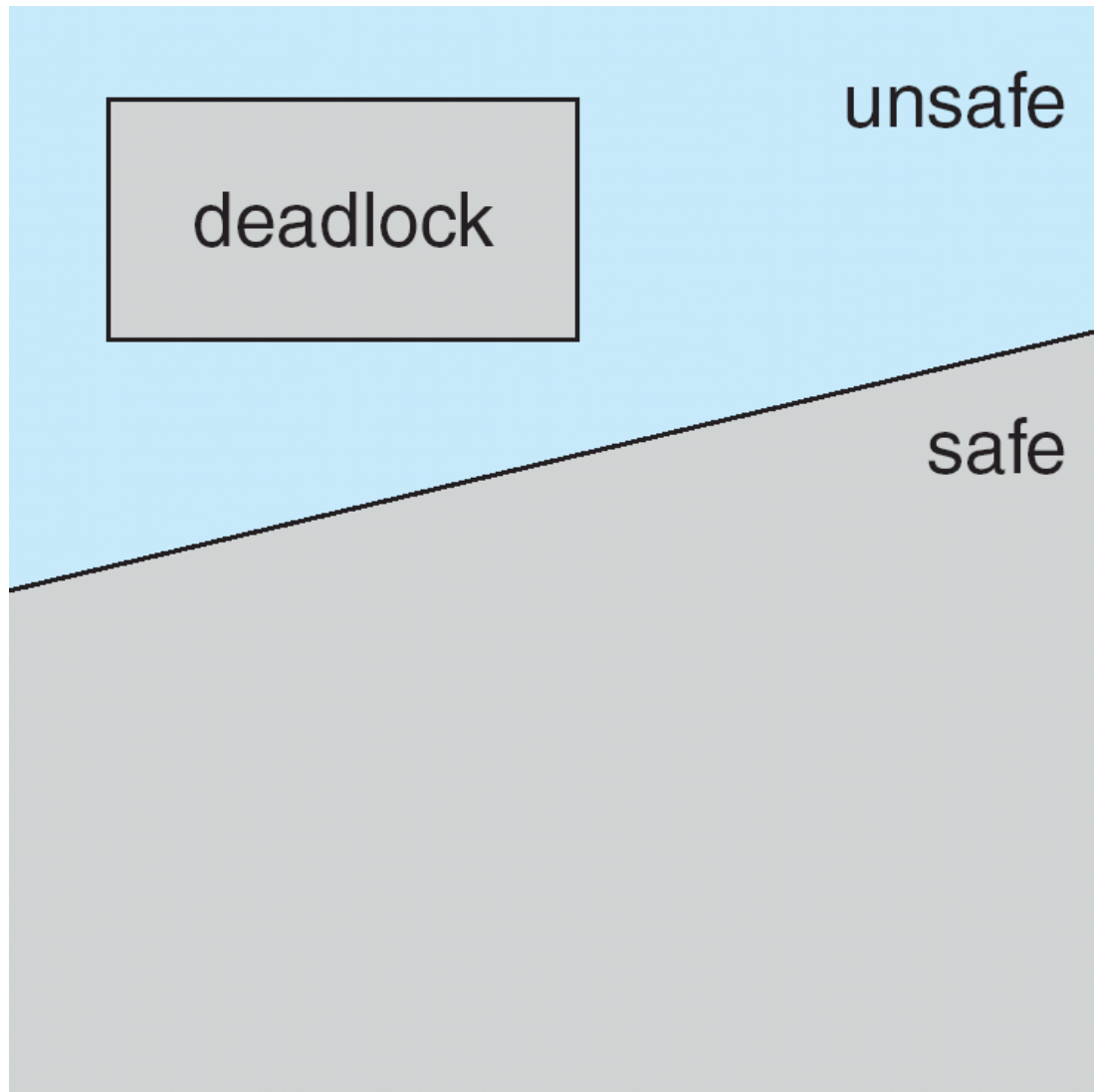
Safe State

- That is:
 - If P_i resource needs are not immediately available, then P_i can wait until all P_j have finished.
 - When P_j is finished, P_i can obtain needed resources, execute, return allocated resources, and terminate.
 - When P_i terminates, P_{i+1} can obtain its needed resources, and so on.

Basic Facts

- If a system is in safe state $\Rightarrow$ no deadlocks
- If a system is in unsafe state $\Rightarrow$ possibility of deadlock
- Avoidance $\Rightarrow$ ensure that a system will never enter an unsafe state.

Safe, Unsafe , Deadlock State



Avoidance algorithms

- Single instance of a resource type
 - Use a resource-allocation graph
- Multiple instances of a resource type
 - Use the banker's algorithm

Banker's Algorithm

- **Multiple instances**

When a new process enters the system, it must declare the maximum number of instances of each resource type that it may need.

- This number may not exceed the total number of resources in the system.
- When a user requests a set of resources, the system must determine whether the allocation of these resources will leave the system in a safe state.
- If it will, the resources are allocated; otherwise, the process must wait until some other process releases enough resources.

Data Structures for the Banker's Algorithm

Let n = number of processes, and m = number of resources types.

- **Available:** Vector of length m . If available $[j] = k$, there are k instances of resource type R_j available
- **Max:** $n \times m$ matrix. If $Max[i,j] = k$, then process P_i may request at most k instances of resource type R_j
- **Allocation:** $n \times m$ matrix. If $Allocation[i,j] = k$ then P_i is currently allocated k instances of R_j
- **Need:** $n \times m$ matrix. If $Need[i,j] = k$, then P_i may need k more instances of R_j to complete its task

$$Need[i,j] = Max[i,j] - Allocation[i,j]$$

Safety Algorithm

1. Let *Work* and *Finish* be vectors of length m and n , respectively. Initialize:
 $\text{Work} = \text{Available}$
 $\text{Finish}[i] = \text{false}$ for $i = 0, 1, \dots, n-1$
2. Find an i such that both:
 - (a) $\text{Finish}[i] = \text{false}$
 - (b) $\text{Need}_i \leq \text{Work}$If no such i exists, go to step 4
3. $\text{Work} = \text{Work} + \text{Allocation}_i$
 $\text{Finish}[i] = \text{true}$
go to step 2
4. If $\text{Finish}[i] == \text{true}$ for all i , then the system is in a safe state

Resource-Request Algorithm for Process P_i

Request = request vector for process P_i . If $Request_i[j] = k$ then process P_i wants k instances of resource type R_j

1. If $Request_i \leq Need_i$ go to step 2. Otherwise, raise error condition, since process has exceeded its maximum claim
2. If $Request_i \leq Available$, go to step 3. Otherwise P_i must wait, since resources are not available
3. Pretend to allocate requested resources to P_i by modifying the state as follows:

$$Available = Available - Request;$$

$$Allocation_i = Allocation_i + Request_i;$$

$$Need_i = Need_i - Request_i;$$

- If safe $\Rightarrow$ the resources are allocated to P_i
- If unsafe $\Rightarrow P_i$ must wait, and the old resource-allocation state is restored

Example of Banker's Algorithm

- 5 processes P_0 through P_4 ;

3 resource types:

A (10 instances), B (5 instances), and C (7 instances)

Snapshot at time T_0 :

	<u>Allocation</u>			<u>Max Available</u>					
	A	B	C	A	B	C	A	B	C
P_0	0	1	0	7	5	3	3	3	2
P_1	2	0	0	3	2	2			
P_2	3	0	2	9	0	2			
P_3	2	1	1	2	2	2			
P_4	0	0	2	4	3	3			

Example (Cont.)

- The content of the matrix *Need* is defined to be *Max – Allocation*

	<u><i>Need</i></u>		
	<i>A</i>	<i>B</i>	<i>C</i>
P_0	7	4	3
P_1	1	2	2
P_2	6	0	0
P_3	0	1	1
P_4	4	3	1

- The system is in a safe state since the sequence $\langle P_1, P_3, P_4, P_2, P_0 \rangle$ satisfies safety criteria

Example: P_1 Request (1,0,2)

- Check that Request $\leq$ Available (that is, $(1,0,2) \leq (3,3,2) \Rightarrow$ true

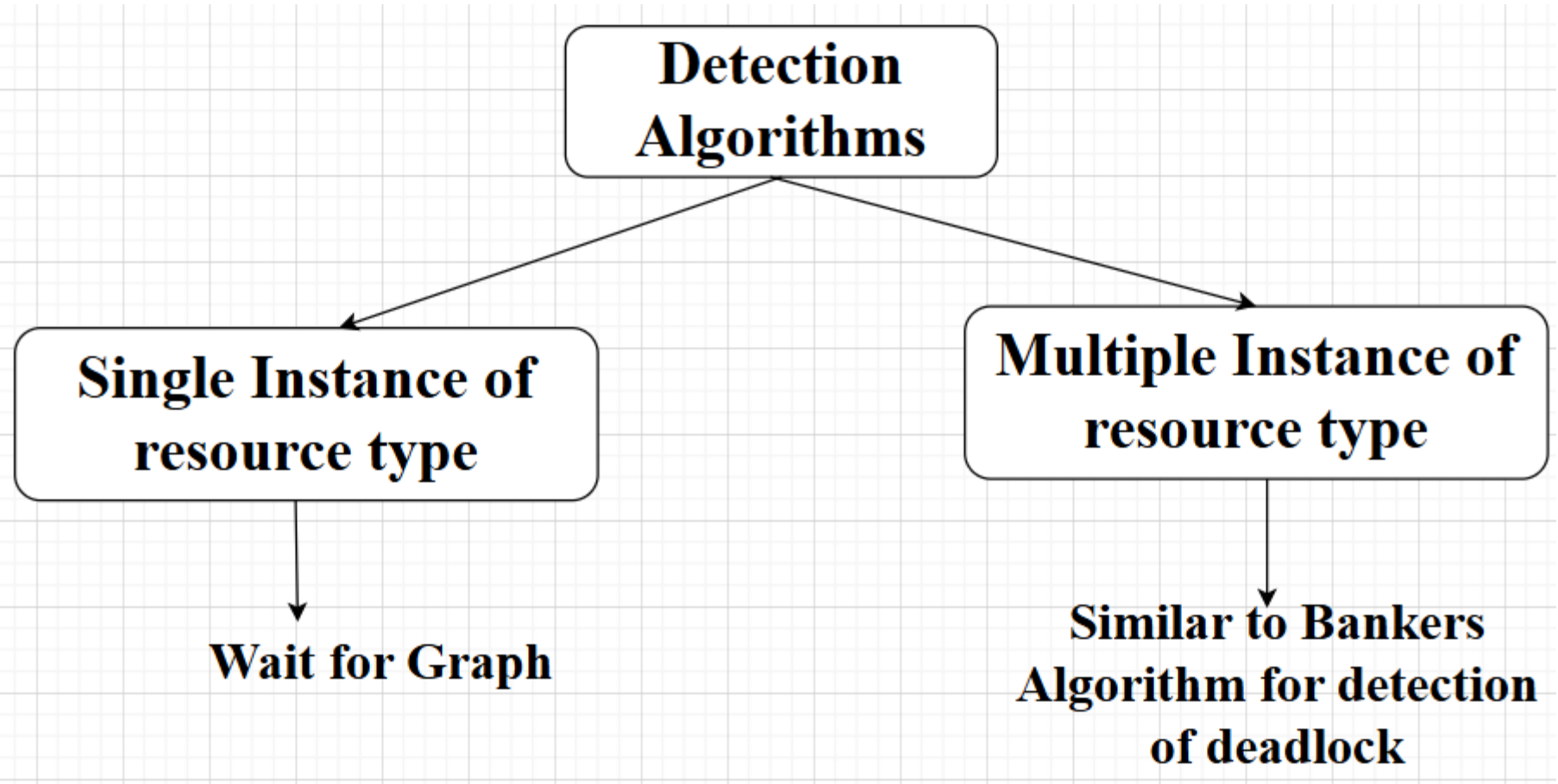
	<u>Allocation</u>	<u>Need</u>	<u>Available</u>
	A B C	A B C	A B C
P_0	0 1 0	7 4 3	2 3 0
P_1	3 0 2	0 2 0	
P_2	3 0 2	6 0 0	
P_3	2 1 1	0 1 1	
P_4	0 0 2	4 3 1	

- Executing safety algorithm shows that sequence $\langle P_1, P_3, P_4, P_0, P_2 \rangle$ satisfies safety requirement
- Can request for (3,3,0) by P_4 be granted?
- Can request for (0,2,0) by P_0 be granted?

Deadlock Detection

- Allow system to enter deadlock state
- Detection algorithm
- Recovery scheme

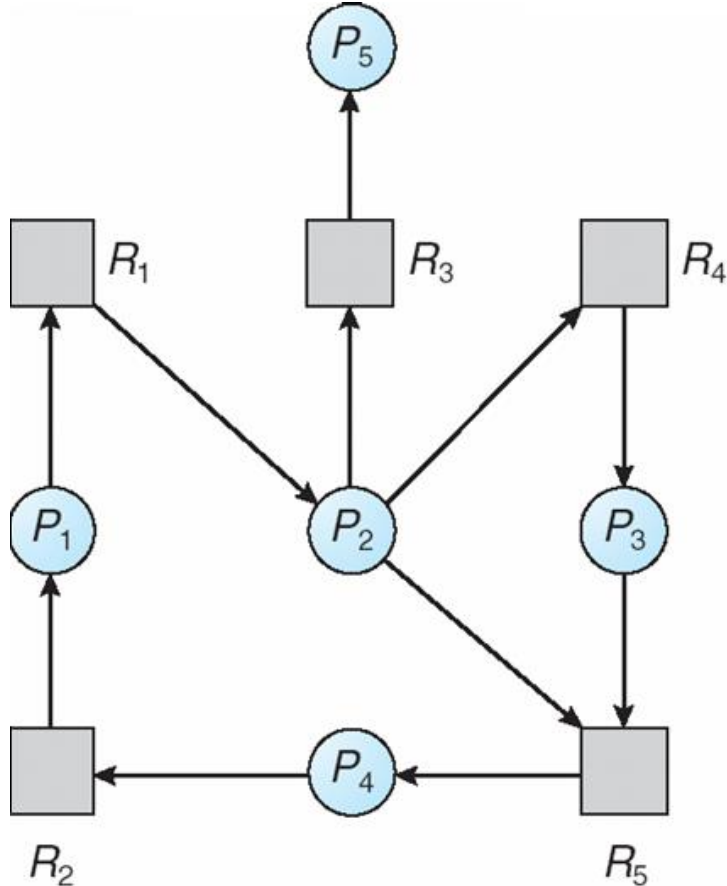
Deadlock Detection



Single Instance of Each Resource Type

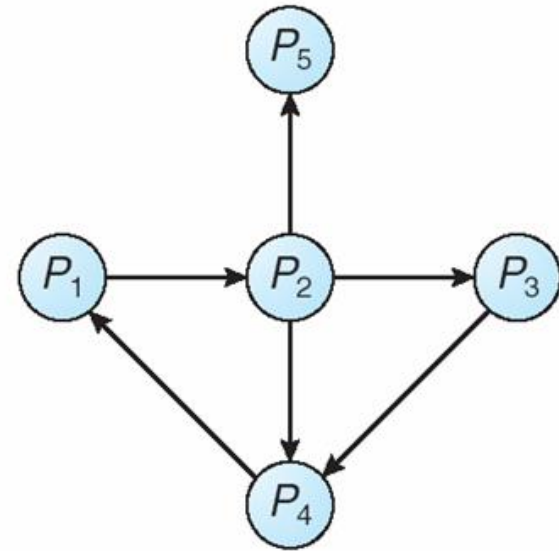
- Maintain *wait-for* graph
 - Nodes are processes
 - $P_i \rightarrow P_j$ if P_i is waiting for P_j
- Periodically invoke an algorithm that searches for a cycle in the graph. If there is a cycle, there exists a deadlock
- An algorithm to detect a cycle in a graph requires an order of n^2 operations, where n is the number of vertices in the graph

Resource-Allocation Graph and Wait-for Graph



(a)

Resource-Allocation
Graph



(b)

Corresponding wait-for graph

Multiple Instances of a Resource Type

- **Available:** A vector of length m indicates the number of available resources of each type.
- **Allocation:** An $n \times m$ matrix defines the number of resources of each type currently allocated to each process.
- **Request:** An $n \times m$ matrix indicates the current request of each process. If $Request[i_j] = k$, then process P_i is requesting k more instances of resource type. R_j .

Detection Algorithm

1. Let *Work* and *Finish* be vectors of length m and n , respectively Initialize:

(a) $Work = Available$

(b) For $i = 1, 2, \dots, n$, if $Allocation_i \neq 0$, then
 $Finish[i] = false$;

otherwise,

$Finish[i] = true$

2. Find an index i such that both:

(a) $Finish[i] == false$

(b) $Request_i \leq Work$

If no such i exists, go to step 4

Detection Algorithm (Cont.)

3. $Work = Work + Allocation_i$
 $Finish[i] = true$
go to step 2
4. If $Finish[i] == false$, for some i , $1 \leq i \leq n$, then the system is in deadlock state. Moreover, if $Finish[i] == false$, then P_i is deadlocked

Algorithm requires an order of $O(m \times n^2)$ operations to detect whether the system is in deadlocked state

Example of Detection Algorithm

- Five processes P_0 through P_4 ; three resource types A (7 instances), B (2 instances), and C (6 instances)
- Snapshot at time T_0 :

	<u>Allocation</u>			<u>Request</u>			<u>Available</u>		
	A	B	C	A	B	C	A	B	C
P_0	0	1	0	0	0	0	0	0	0
P_1	2	0	0	2	0	2			
P_2	3	0	3	0	0	0			
P_3	2	1	1	1	0	0			
P_4	0	0	2	0	0	2			

- Sequence $\langle P_0, P_2, P_3, P_4, P_1, \rangle$ will result in $Finish[i] = \text{true}$ for all i

Example (Cont.)

- P_2 requests an additional instance of type C

Request

$A \ B \ C$

P_0 0 0 0

P_1 2 0 2

P_2 0 0 1

P_3 1 0 0

P_4 0 0 2

- State of system?
 - Can reclaim resources held by process P_0 , but insufficient resources to fulfill other processes; requests
 - Deadlock exists, consisting of processes P_1 , P_2 , P_3 , and P_4

Detection-Algorithm Usage

- When, and how often, to invoke depends on:
 - How often a deadlock is likely to occur?
 - How many processes will need to be rolled back?
 - one for each disjoint cycle
- If detection algorithm is invoked arbitrarily, there may be many cycles in the resource graph and so we would not be able to tell which of the many deadlocked processes “caused” the deadlock.

Recovery from Deadlock: Process Termination

- **Abort all deadlocked processes**
- **Abort one process at a time until the deadlock cycle is eliminated**
- In **partial termination** method, abort the process which incur minimum cost. So, in which order should we choose to abort?
 - Priority of the process
 - How long the process has computed, and how much longer to completion
 - How many types and what types of resources the process has used
 - How many more resources the process needs to complete
 - How many processes will need to be terminated

Recovery from Deadlock: Resource Preemption

- Selecting a victim – minimize cost
- Rollback – return to some safe state, restart process for that state
- Starvation – same process may always be picked as victim, include number of rollback in cost factor

1. **Selecting a victim.** Which resources and which processes are to be preempted? As in process termination, we must determine the order of preemption to minimize cost. Cost factors may include such parameters as the number of resources a deadlocked process is holding and the amount of time the process has thus far consumed.
2. **Rollback.** If we preempt a resource from a process, what should be done with that process? Clearly, it cannot continue with its normal execution; it is missing some needed resource. We must roll back the process to some safe state and restart it from that state.

Since, in general, it is difficult to determine what a safe state is, the simplest solution is a total rollback: abort the process and then restart it. Although it is more effective to roll back the process only as far as necessary to break the deadlock, this method requires the system to keep more information about the state of all running processes.

3. **Starvation.** How do we ensure that starvation will not occur? That is, how can we guarantee that resources will not always be preempted from the same process?

In a system where victim selection is based primarily on cost factors, it may happen that the same process is always picked as a victim. As a result, this process never completes its designated task, a starvation situation any practical system must address. Clearly, we must ensure that a process can be picked as a victim only a (small) finite number of times. The most common solution is to include the number of rollbacks in the cost factor.

Feature	Deadlock Avoidance (Banker's Algorithm)	Deadlock Detection
When it acts	Before deadlock occurs	After deadlock occurs
Strategy	Ensures system remains in a safe state	Periodically checks for circular wait conditions
Resource allocation	Prevents allocation if it leads to an unsafe state	Allocates resources freely, but detects deadlocks later
System overhead	Higher, as it requires checking safety state before every allocation	Lower, as it only checks periodically

End of the Chapter